OPERATIONS · EDGE CASES · 1/10

Users don't read your spec.

What actually breaks in production when real people meet perfect software.

10 slides · swipe

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The spec describes the ideal user.

The ideal user fills every field, speaks one language, pays attention and never retries. The actual user is three time zones away, mid-migraine, with a typo and a deadline.

Design for the person who just wants it to work.

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What Safalife taught me about edge cases.

01 Users leave address fields blank — landmarks instead

↓

02 Phone numbers arrive half-typed, Russian and Uzbek mixed

↓

03 Dispatchers edit orders mid-route, humans override apps

↓

04 Every 'impossible input' became a support ticket by Tuesday

The chaos field report beats any user research deck.

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The top five production edge cases.

01 Empty required fields that somehow shipped

02 Unicode chaos — mixed scripts, emoji, invisible chars

03 Retry storms — same order placed nine times

04 Clock skew — timestamps from the future

05 Users who close the tab mid-mutation

Every startup's 'weird outage' is one of these five.

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Design for the edge, not the center.

01 Validate at the boundary — reject loudly, don't guess



02 Make mutations idempotent — retries must be safe



03 Give humans override rights — they'll fix it anyway



04 Log everything — the edge cases are your roadmap

Half of reliability is expecting the unexpected — and surviving it.

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Two user models.

FICTION

The spec user

fills every field
reads the docs
patient, complete

PRODUCTION

The 3 a.m. user

half-awake
guessed the address
needs delivery now

Your API has to survive the second one.

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The edge-case audit for your product.

01 What happens on empty input at every write path?

02 What happens on a duplicated submission?

03 What happens when the tab closes mid-write?

04 What happens when the clock is wrong?

If 'it depends' is the answer to two of these, you have work to do.

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What designing for edge cases buys.

Fewer silent corruptions. Fewer angry calls. A database that reflects reality — and a support team that can actually help instead of apologizing.

**Every edge case survived
is a support ticket that never happened.**

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The edge case law.

If your system can't describe what the user will do wrong, it can't survive what the user will do.

The messy middle is the product. The spec is just the brochure.

Save this for your next user review.

Next time someone says 'users won't do that' — hand them this deck. The users already did. They left it in your logs Tuesday.

Save for later review

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